Module 2 :

Scheduling Algorithm

• **Non-preemptive**	• **Preemptive**
1. FCFS	1. SRTN
2. SJF	2. Round-robin (RR)
3. Priority based	3. Priority based

Numericals:-- Non Premptive TYPE

1)Consider the following snapshot of the process to be executed. Draw the Gantt chart and **determine the average waiting time and average turnaround time** for SJF (NON- preemptive),

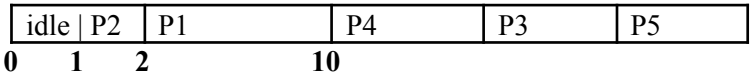
FCFS and Priority scheduling (non-preemptive) scheduling algorithm.

a) **SJF (criteria :--BT)**

Process	Arrival Time	Burst Time
P1	0	8
P2	1	1
P3	2	3
P4	3	2
P5	4	6

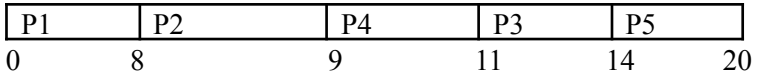
Process	Arrival Time	Burst Time
P1	1	8
P2	1	1

P3	3	3
P4	4	2
P5	5	6



Process	Arrival Time	Burst Time
P1	0	8
P2	1	1
P3	2	3
P4	3	2
P5	4	6

GANN CHART:



Process	Arrival Time	Burst Time	CT	TAT (CT-AT)	WT (TAT-BT)
P1	0	8	8	8	0
P2	1	1	9	8	7
P3	2	3	14	12	9
P4	3	2	11	8	6
P5	4	6	20	16	10

Avg WT = (0+7+9+6+10)/5 = 32/5= 6.4

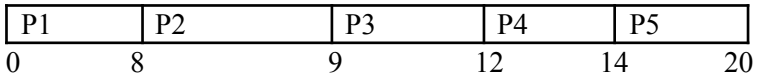
Avg TAT = 10.4

b) FCFS (criterion :- AT) Non-preemptive

Process	Arrival Time	Burst Time

P1	0	8
P2	1	1
P3	2	3
P4	3	2
P5	4	6

GANN CHART:



Process	Arrival Time	Burst Time	CT	TAT (CT-AT)	WT (TAT-BT)
P1	0	8	8	8	0
P2	1	1	9	8	7
P3	2	3	12	10	7
P4	3	2	14	11	9
P5	4	6	20	16	10

Avg WT = (0+7+7+9+10)/5 = 33/5= **6.6**

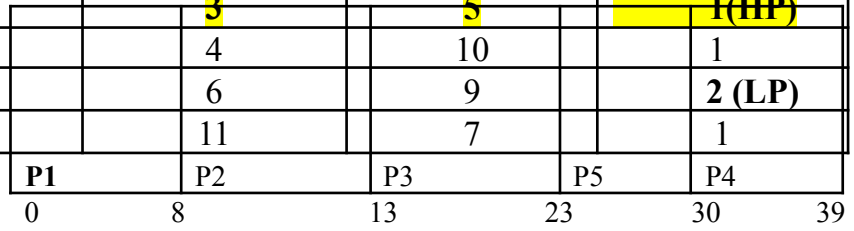
Avg TAT = **10.6**

C) Priority scheduling (non premptive) :--

- process with **same priority are executed in FCFS manner (considering AT)**
  process having less value= highest priority

Process	Arrival Time	Burst Time	Priority
P1	0	8	2
P2	**3**	**5**	**1(HP)**
P3	4	10	1
P4	6	9	**2 (LP)**
P5	11	7	1

GANN CHART:



Process	Arrival Time	Burst Time	Priority	CT	TAT (CT-AT)	WT (TAT-BT)
P1	0	8	2	8	8	0
P2	3	5	1	13	10	5
P3	4	10	1	23	19	9
P4	6	9	2	39	33	24
P5	11	7	1	30	19	12

Avg WT = 10

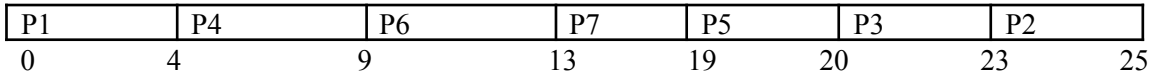
Avg TAT = 17.8

D) Priority scheduling :--

- High value= high priority
- process with same priority are executed in FCFS manner

Process	Arrival Time	Burst Time	Priority
P1	0	4	**2(L)**
P2	1	2	4
P3	2	3	6
P4	3	5	10
P5	4	1	8
P6	**5**	4	**12(H)**
P7	6	6	9

GANN CHART:



Process	Arrival Time	Burst Time	Priority	CT	TAT (CT-AT)	WT (TAT-BT)
P1	0	4	2(L)			
P2	1	2	4			
P3	2	3	6			
P4	3	5	10			
P5	4	1	8			
P6	5	4	12(H)			
P7	6	6	9			

- **For P4:-- till AT=4 , we need to see all processes who has highest priority, so here P4 has highest priority.**
- **When A.T completes till 9, A.t=9 all processes are available so place all process now in FCFS manner.**

Avg WT =

Avg TAT =

Prememptive Types:--

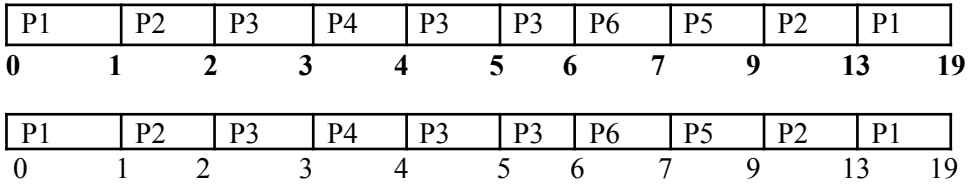
A) Shortest Remaining time first (SRTF) (Criteria- Burst time, Quantum time= 1sec)

Process	Arrival Time	Burst Time	CT	TAT	WT
P1	0	7 -1=6	19	19	
P2	1	5 -1=4	13	12	
P3	2	3 -1=2-1=1,0	6	4	
P4	3	1 -1=0	4	1	
P5	4	2	9	5	
P6	5	1	7	2	

Process	Arrival Time	Burst Time	CT	TAT	WT

P1	0	7 -1=6	19	19	
P2	1	5 -1=4	13	12	
P3	2	3 -1=2-1=1-1=0	6	4	
P4	3	1 -1=0	4	1	
P5	4	2	9	5	
P6	5	1	7	2	

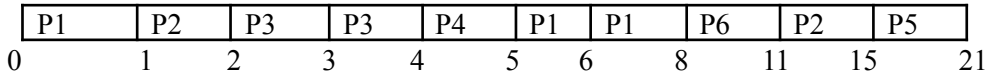
GANN CHART:



Avg W.T==

Process	Arrival Time	Burst Time
P1	0	4-1=3-1=2,0
P2	1	5-1=4
P3	2	2-1=1,0
P4	3	1,0
P5	4	6
P6	6	3

GANN CHART:

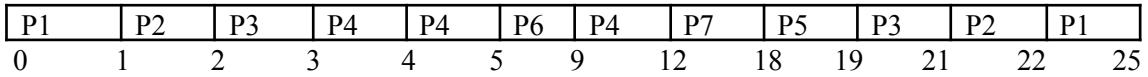


Avg TAT =

B) Priority scheduling :-- (Premptive) QT=1sec

Process	Arrival Time	Burst Time	Priority
P1	0	4-1=3	2(L)
P2	1	2-1=1	4
P3	2	3-1=2	6
P4	3	5-1=4-1=3	10
P5	4	1	8
P6	5	4-1=3	12(H)
P7	6	6	9

GANN CHART:



Process	Arrival Time	Priority	Burst Time	CT	TAT (CT-AT)	WT (TAT-BT)
P1	0	2(L)	4	25		
P2	1	4	2	22		
P3	2	6	3	21		
P4	3	10	5	12		
P5	4	8	1	19		
P6	5	12(H)	4	09		
P7	6	9	6	18		

Here P1 will run first as AT= 0 for 1 QT.

Next P2 will run for 1 QT and next P3 will arrive for 1 QT

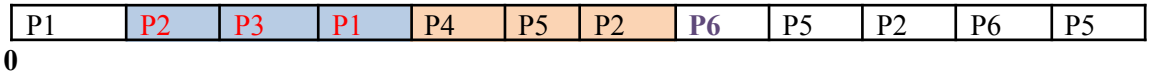
At AT=3 ,P4 will run

Next after P4,P6 has highest priority so P6 will arrive & will run for 4 units.

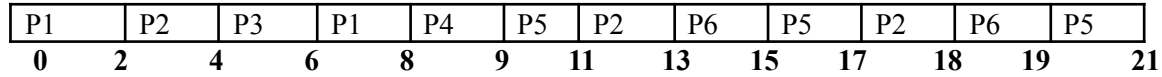
C) Round Robin (criteria :: QT+AT) QT=2

Process	Arrival Time	Burst Time	
P1	0	4-2=2-2=0	
P2	1	5-2=3-2=1	
P3	2	2-2=0	
P4	3	1-1=0	
P5	4	6-2=4-2=2	
P6	6	3-2=1	

Queue:



Gann chart:--



1) Till QT=2, we have p2 ,p3 till time 2. P1 has to go out from queue
2) Next P1 will be in queue, & p2 will execute for 2 units.
3) Till 4 units P4, & P5 already arrived .As P2 not completed process, it should be in queue after P4 & P4 .
4) By time 6 units, process P6 already arrived.